# Sep 8, 2026 (Due: 08:00 Sep 15, 2026)

**1.** Let $L \in \mathbb{R}^{n \times n}$ be lower triangular and nonsingular. Provide two different implementations for solving the linear system $Ux = b$, where $b \in \mathbb{R}^n$ is a given vector.

**2.** Let $L \in \mathbb{R}^{n \times n}$ be lower triangular and nonsingular. Design an algorithm to compute $L^{-1}$, which overwrites $L$. Make sure your algorithm only accesses the lower triangular entries of $L$ and $O(1)$ auxiliary space.

**3.** Let $A$ be a nonsingular banded matrix with half-bandwidth $b$ (i.e., $a_{i,j} = 0$ if $|i-j| > b$). Suppose that the LU factorization of $A$ (without pivoting) is $A = LU$. Show that both $L$ and $U$ are banded matrices with half-bandwidth $b$.

**4.** Find the exact LU factorization of the $n \times n$ matrix

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 1 \\ -1 & 1 & 0 & \cdots & 0 & 1 \\ -1 & -1 & 1 & \cdots & 0 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -1 & -1 & \cdots & 1 & 1 \\ -1 & -1 & -1 & \cdots & -1 & 1 \end{bmatrix}.$$

**5.** Implement Gaussian elimination (without pivoting) for solving nonsingular linear systems. You may assume that no divide-by-zero error is encountered. Measure the execution time of your program in terms of matrix dimensions and visualize the result by a log–log scale plot. (You may generate your test matrices with with normally distributed random elements.)

**6.** (optional) Let $U_1, U_2 \in \mathbb{C}^{n \times n}$ be upper triangular, $\mu \in \mathbb{C}$, and $b \in \mathbb{C}^n$. Suppose that $\det(U_2 U_1 - \mu I) \neq 0$. Design an $O(n^2)$ algorithm to solve the linear system $(U_2 U_1 - \mu I)x = b$.

**7.** (optional) Suppose that you have a BLAS library with an inefficient implementation of `STRSM`. Try to design an efficient `STRSM` subroutine by making use of `SGEMM`. You do not have to really implement it. Just describe your algorithm.